

Translated

From: e-maxx.ru

Balanced Ternary



This is a non-standard but still positional **numeral system**. Its feature is that digits can have one of the values `-1`, `0` and `1`. Nevertheless, its base is still `3` (because there are three possible values). Since it is not convenient to write `-1` as a digit, we'll use letter `Z` further for this purpose. If you think it is quite a strange system - look at the picture - here is one of the computers utilizing it.

So here are few first numbers written in balanced ternary:

0	0
1	1
2	1Z
3	10
4	11
5	1ZZ
6	1Z0
7	1Z1
8	10Z
9	100

This system allows you to write negative values without leading minus sign: you can simply invert digits in any positive number.

-1	Z
-2	Z1
-3	Z0